

HCDE 511

Information Visualization

Winter 2026

1
Week

This course content has been built on the amazing work of
Cecilia Aragon, Brock Craft, and Holger Kuehnle

Week

Introductions

In-class activity

Break

Introduction to information visualization

Syllabus + logistics

Break

Data encoding & mapping

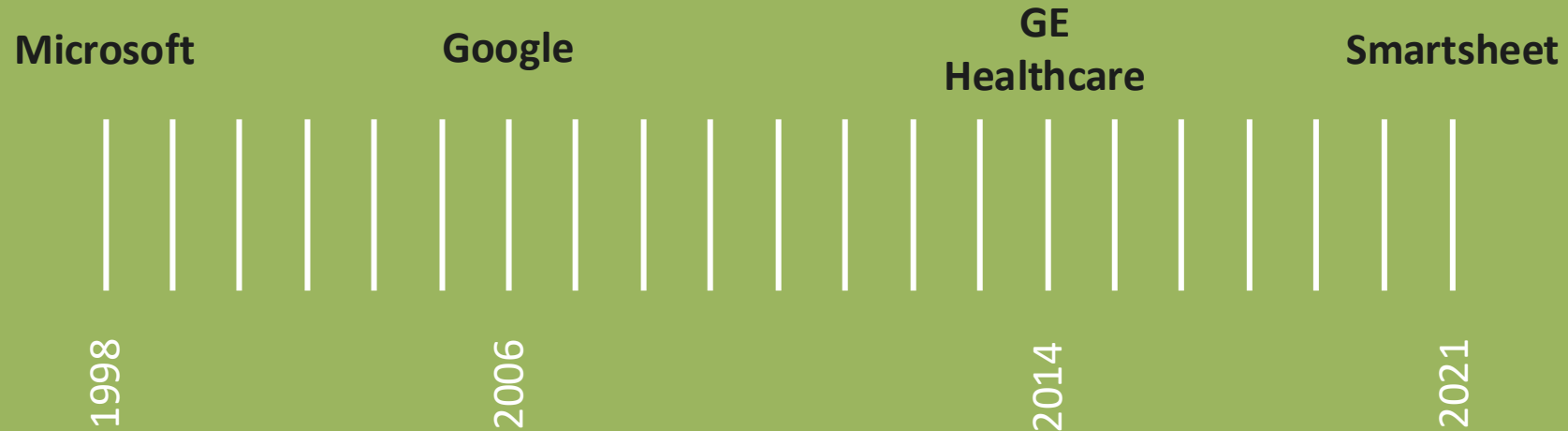
who

Douglas Pyle (he/him)

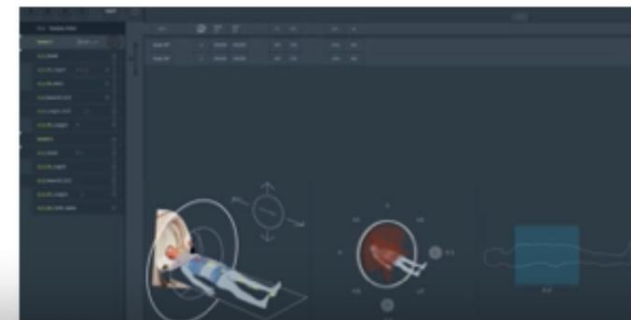
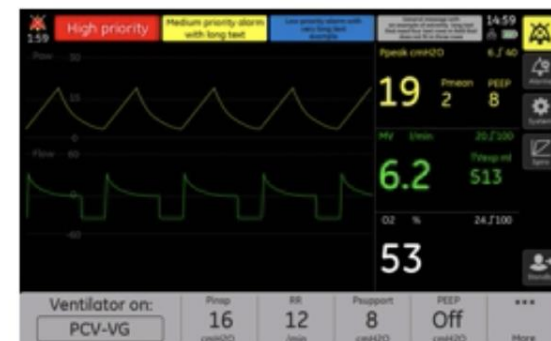
DougDoug@uw.edu

BA, English Literature & Marketing Management

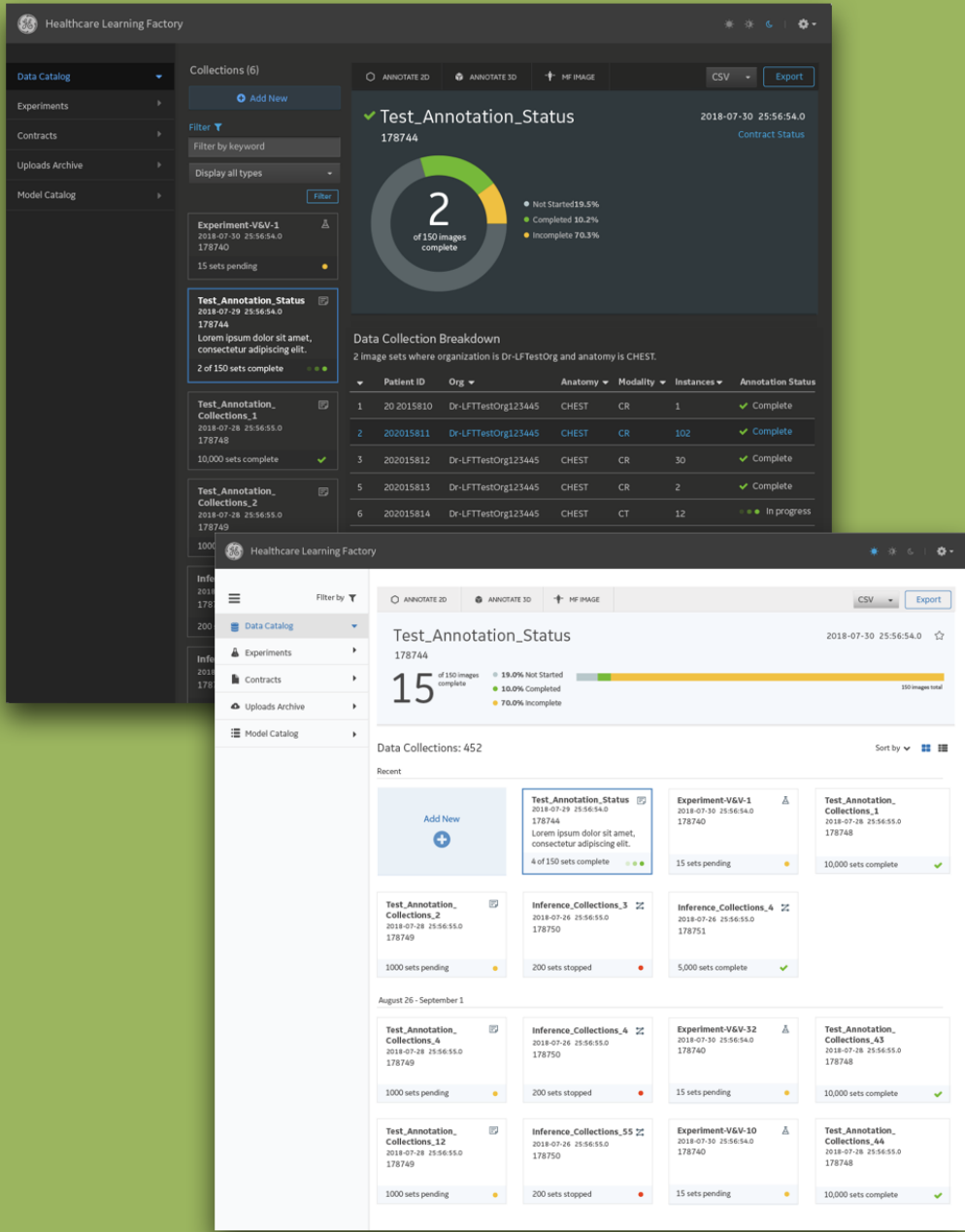
MS, Dept. Human Centered Design & Engineering



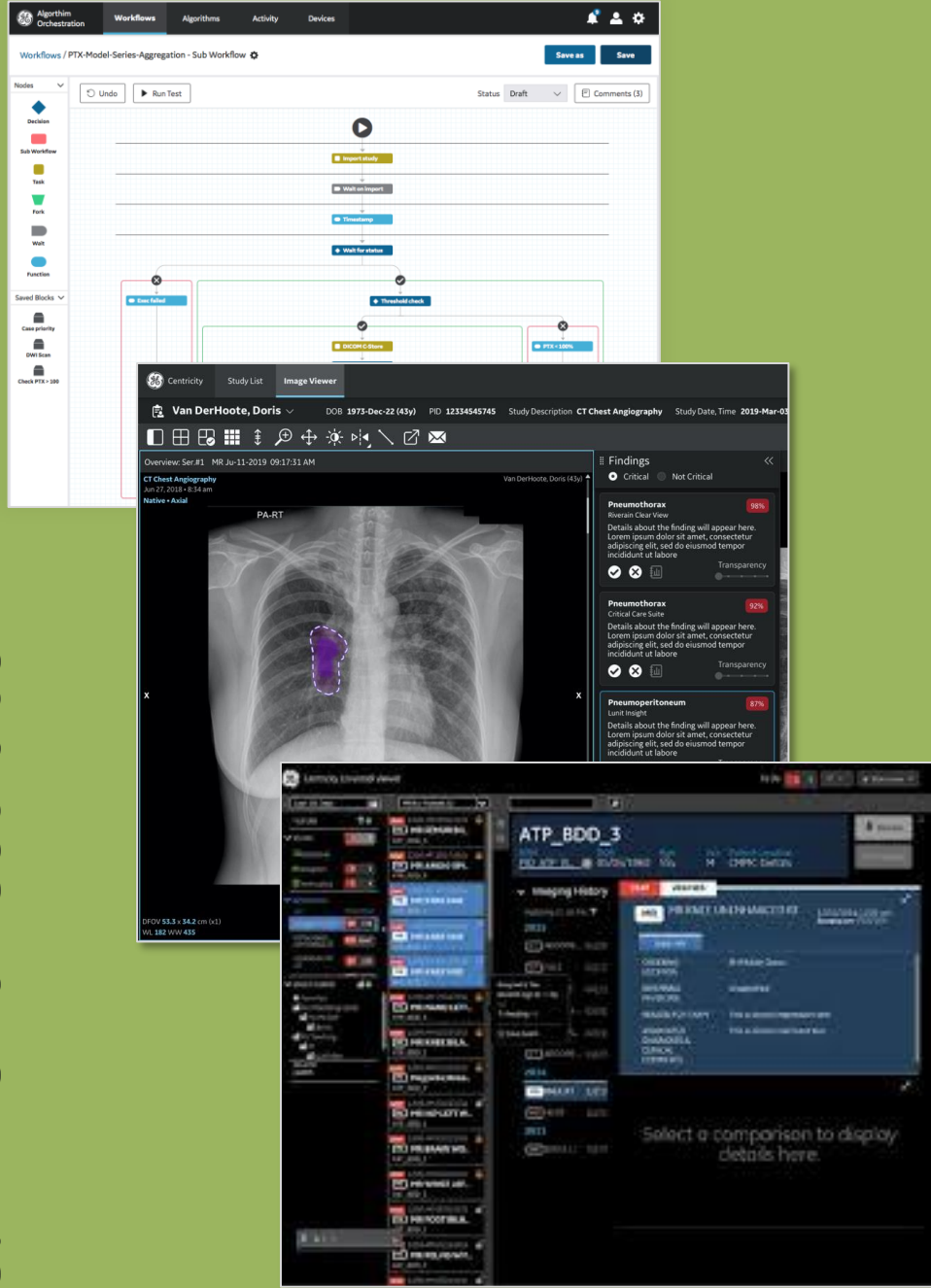
GE Healthcare



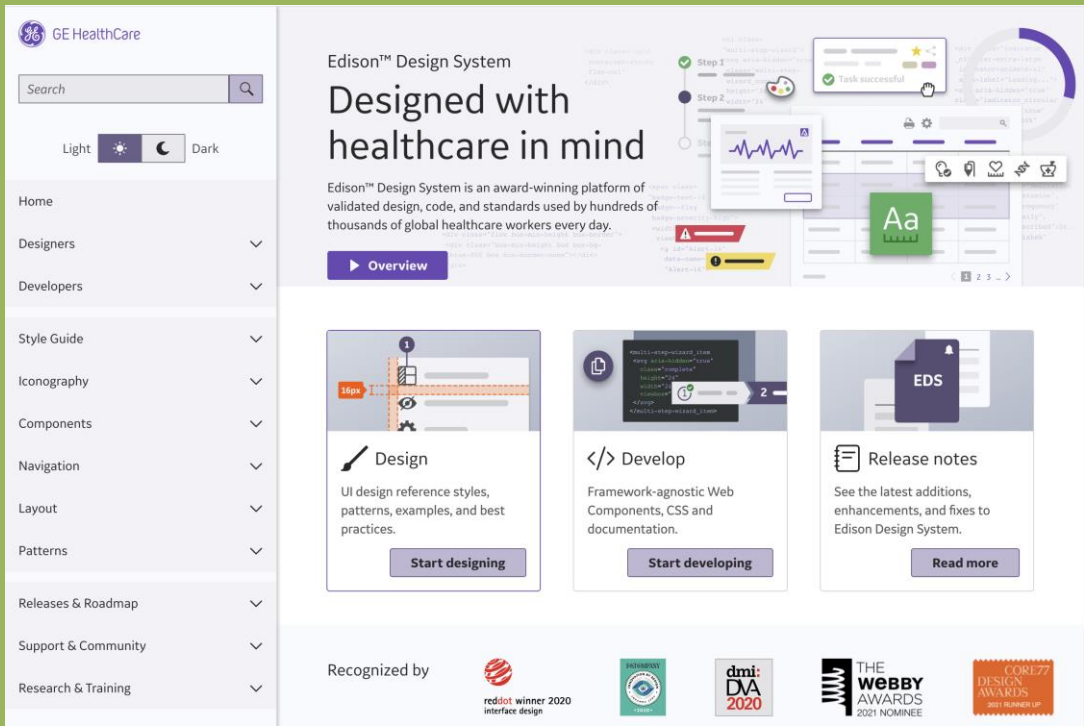
Edison AI Data Science Platform



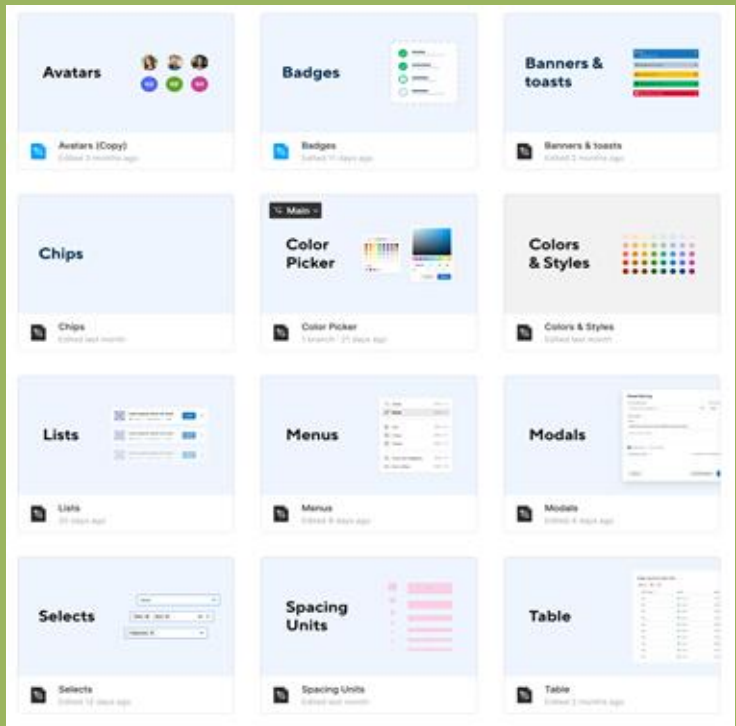
Centricity Workflow Viewer & AI Orchestrator



GE Edison Design System



Smartsheet Design System



Donghoon Shin (dhoon@uw.edu)

<https://donghoon.io>

Donghoon Shin (he/him)

<https://donghoon.io> | dhoon@uw.edu

I am a fourth-year HCDE Ph.D. student working with Gary Hsieh and Lucy Lu Wang.

My research lies at the intersection of HCI and AI, where I build systems that streamline the dissemination, access, and consumption of complex documents (e.g., scholarly papers) to support real-world designing.

I did my bachelor in ECE and Information Science at Seoul National University in Korea. Previously, I worked at Microsoft Research, Google Research, Adobe Research, and Naver AI as a research intern.



**Be sure to complete
the Project Group
Assignment Survey
on Canvas.**

<https://forms.gle/MYPPMfMbHLXNRJV38>

Activity

Examine and explore a dataset

Identify interesting characteristics and relationships

Decide upon a “message” / story you could convey supported by the data

Construct a representation from any materials you can find... ;)

Make something physical! Can use pen + paper for labeling if needed.

Country	Country Code	GDP Rank	GDP 1	GDP (per capita)	Population (2012)	Net migration (2012)	Percent of Individuals using the Internet (2012)	Air	Proportion	Life	International tourism, number of arrivals (2011)
			(millions of US dollars) (2012)	(US dollars) (2012)				transport, registered carrier departures worldwide (2012)	of seats held by women in national parliaments (%) (2012)	expectancy at birth, total (years) (2011)	
United States	USA	1	15,684,800	49,965	313,914,040	4,954,924	81	9,860,451	18	79	62,711,000
China	CHN	2	8,227,103	6,091	1,350,695,000	-1,884,102	42	2,779,741	21	73	57,581,000
Japan	JPN	3	5,959,718	46,720	127,561,489	270,000	79	874,677	8	83	6,219,000
Germany	DEU	4	3,399,589	41,514	81,889,839	550,001	84	1,034,154	33	81	28,374,000
France	FRA	5	2,612,878	39,772	65,696,689	500,001	83	655,359	27	82	81,411,000
United Kingdom	GBR	6	2,435,174	38,514	63,227,526	1,020,211	87	1,035,646	23	81	29,306,000
Brazil	BRA	7	2,252,664	11,340	198,656,019	-499,999	50	1,000,827	9	73	5,433,000
Russia	RUS	8	2,014,775	14,037	143,533,000	1,135,737	53	636,748	14	69	24,932,000
Italy	ITA	9	2,013,263	33,049	60,917,978	1,998,926	58	305,512	21	82	46,119,000
India	IND	10	1,841,717	1,489	1,236,686,732	-2,999,998	13	660,862	11	65	6,309,000
Canada	CAN	11	1,821,424	52,219	34,880,491	1,098,444	87	1,280,198	25	81	16,014,000
Australia	AUS	12	1,520,608	67,036	22,683,600	1,124,639	82	611,472	25	82	5,875,000
Spain	ESP	13	1,349,351	29,195	46,217,961	2,250,005	72	435,141	36	82	56,694,0.00
Mexico	MEX	14	1,177,271	9,742	120,847,477	-1,805,238	38	410,584	37	77	23,403,000
South Korea	KOR	15	1,129,598	22,590	50,004,000	-30,000	84	285,000	16	81	9,795,000
Indonesia	IDN	16	878,043	3,557	246,864,191	-1,293,089	15	528,842	19	69	7,650,000
Turkey	TUR	17	789,257	10,666	73,997,128	-50,000	45	476,115	14	74	34,038,000
Netherlands	NLD	18	772,227	46,054	16,767,705	50,006	93	259,784	39	81	11,300,000
Switzerland	CHE	19	632,194	79,052	7,997,152	182,803	85	259,190	29	83	8,534,000
Saudi Arabia	SAU	20	576,824	20,391	28,287,855	1,055,517	54	208,697	0	74	17,498,000

Examine and explore the dataset

Identify interesting characteristics and relationships

Decide upon a “message” / story you could convey supported by the data

Construct a representation

Do not just illustrate an x/y chart on your tabletop or whiteboard

Show a relationship/trend/novelty

Your manifestation must have a title containing a verb

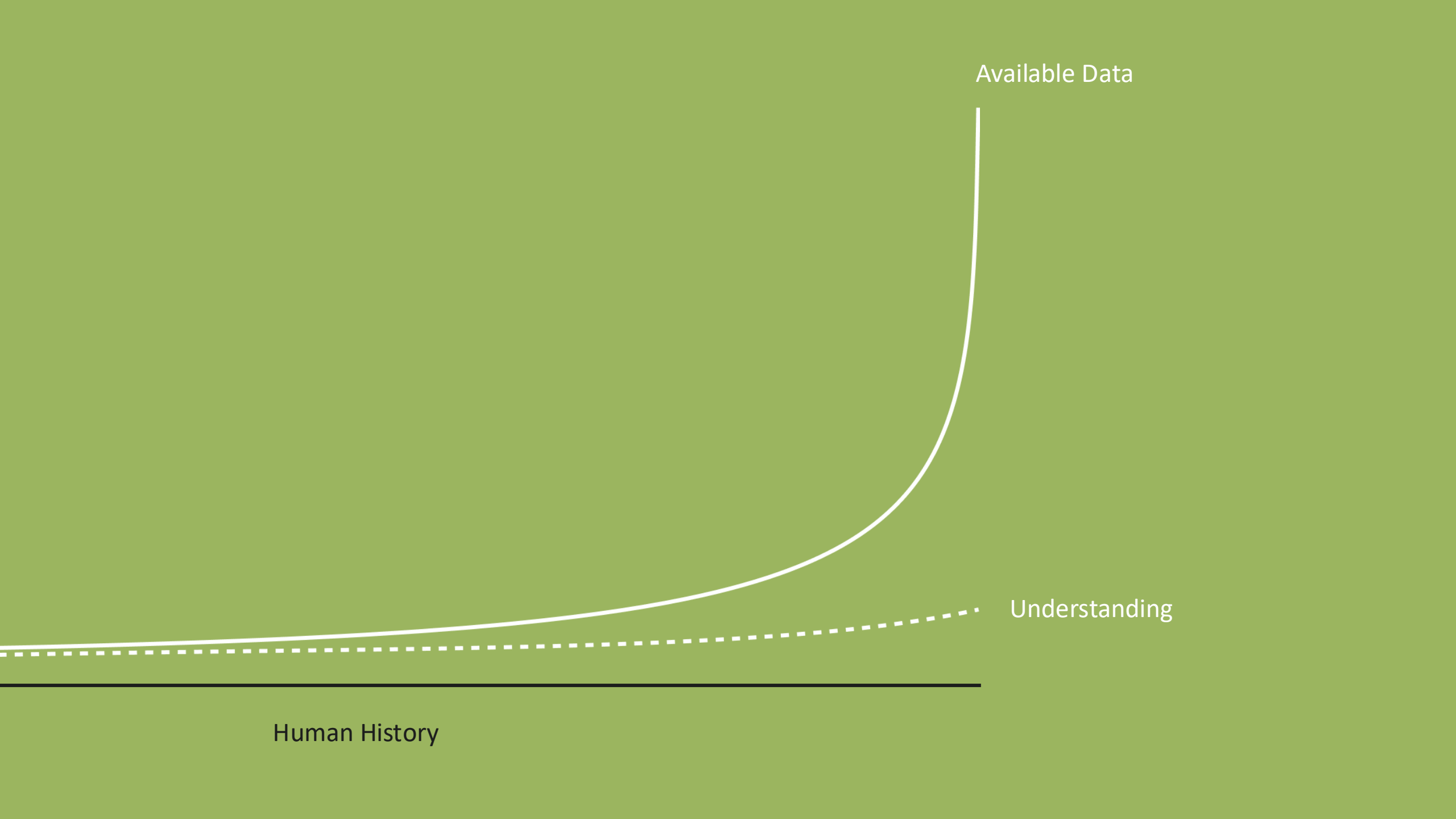
Capture and post it to Canvas before 11:59p

Break

Intro

“What information consumes is rather obvious: It consumes the attention of its recipients. Hence a wealth of information creates a poverty of attention.”

Herbert Simon, September 1995



Available Data

Understanding

Human History

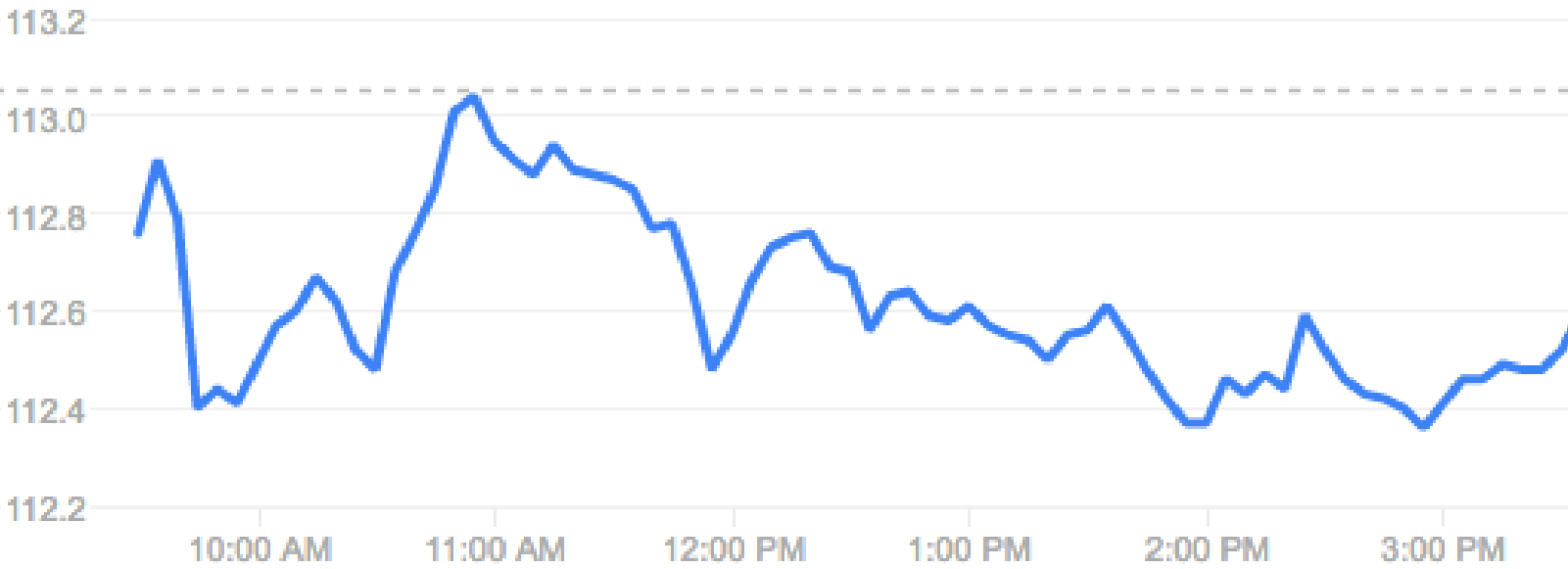




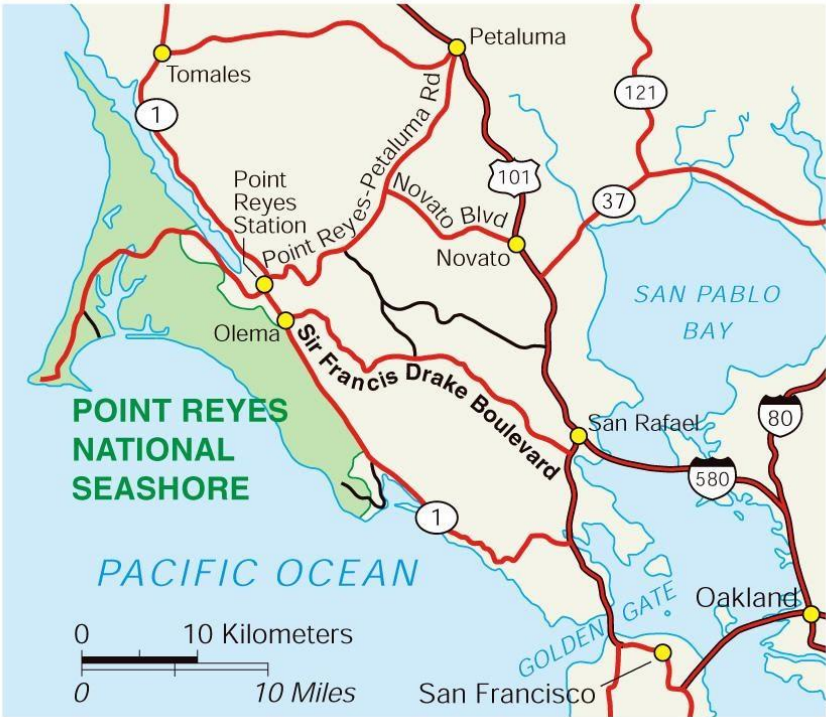
Apple Inc.
NASDAQ: AAPL - Oct 3, 3:59 PM EDT

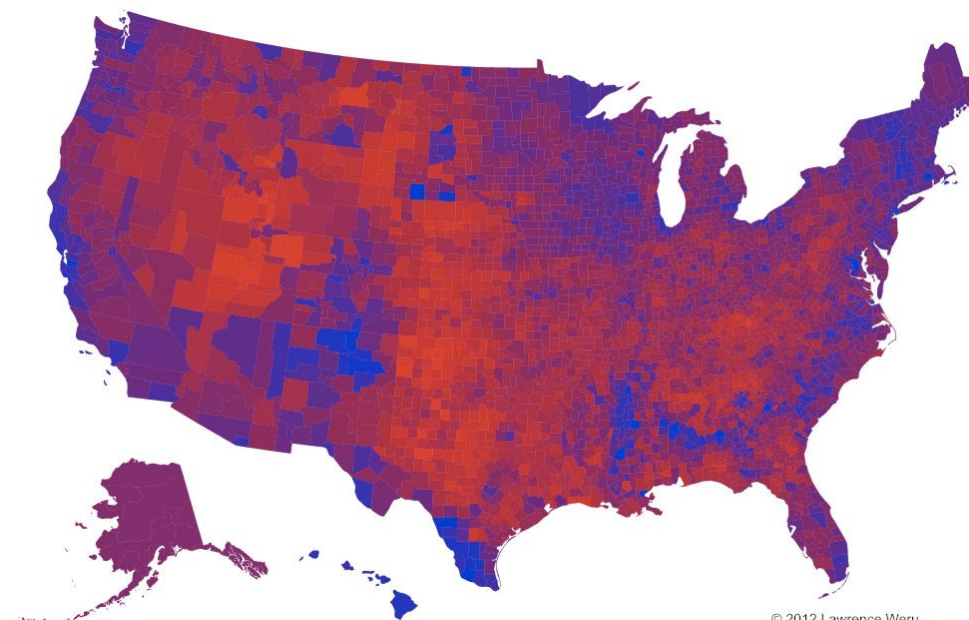
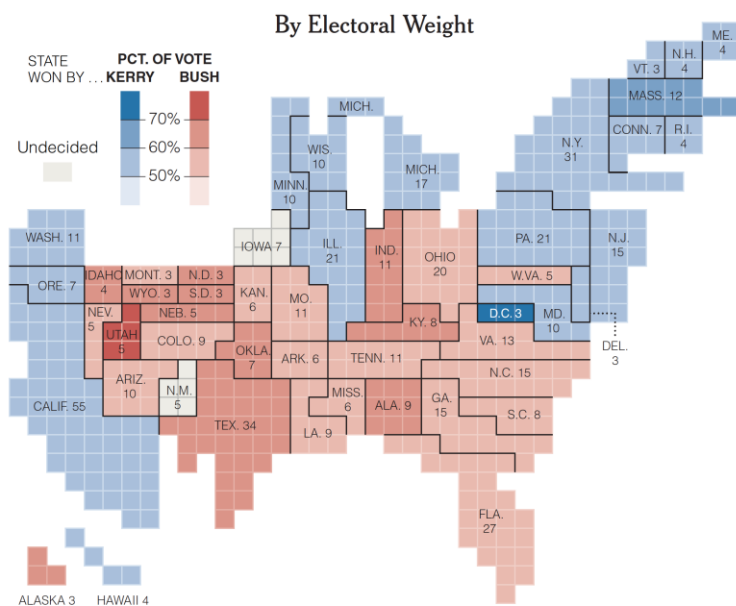
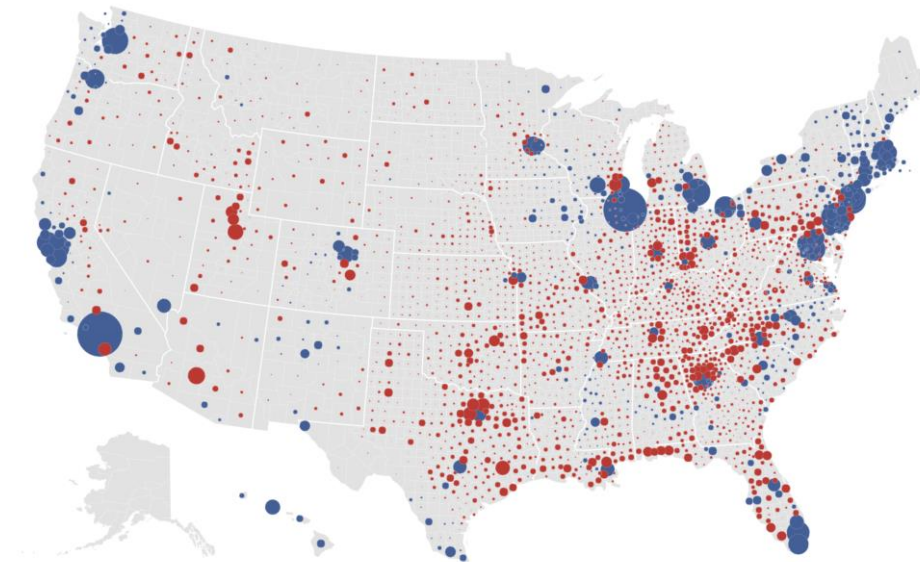
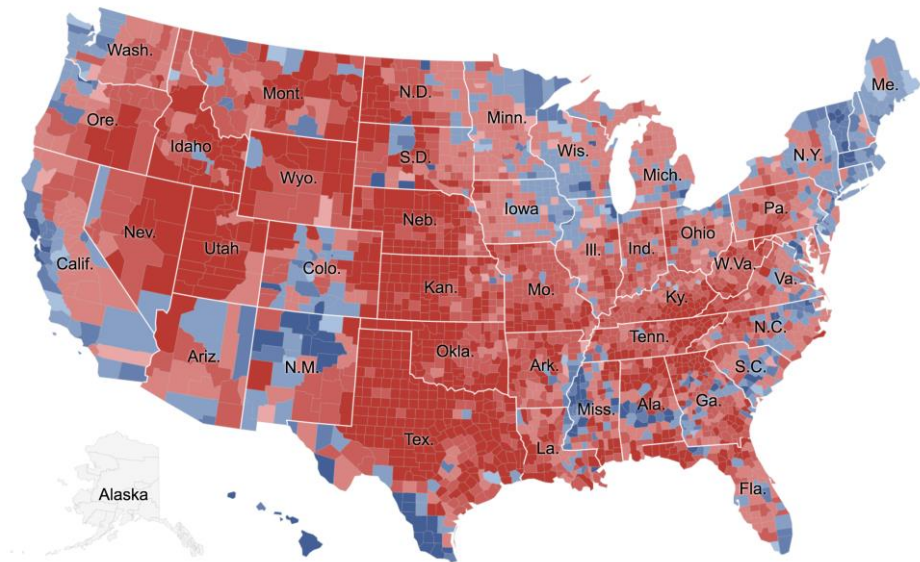
112.53 USD ↓0.52 (0.46%)

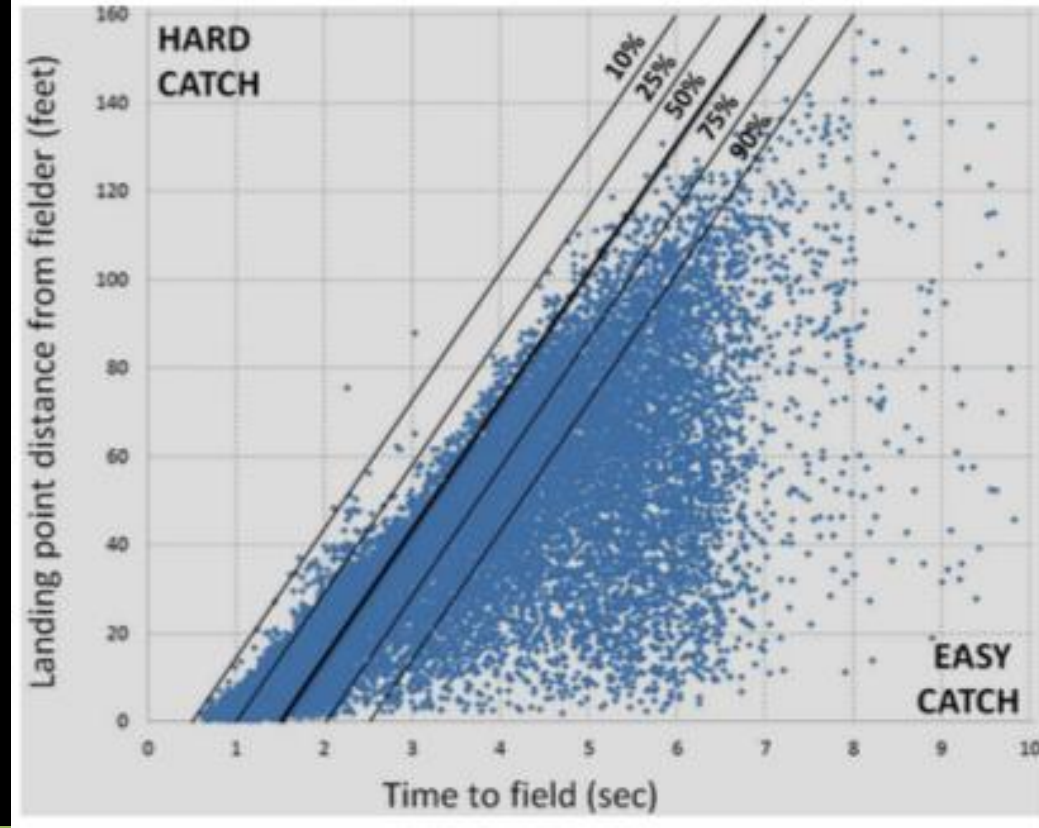
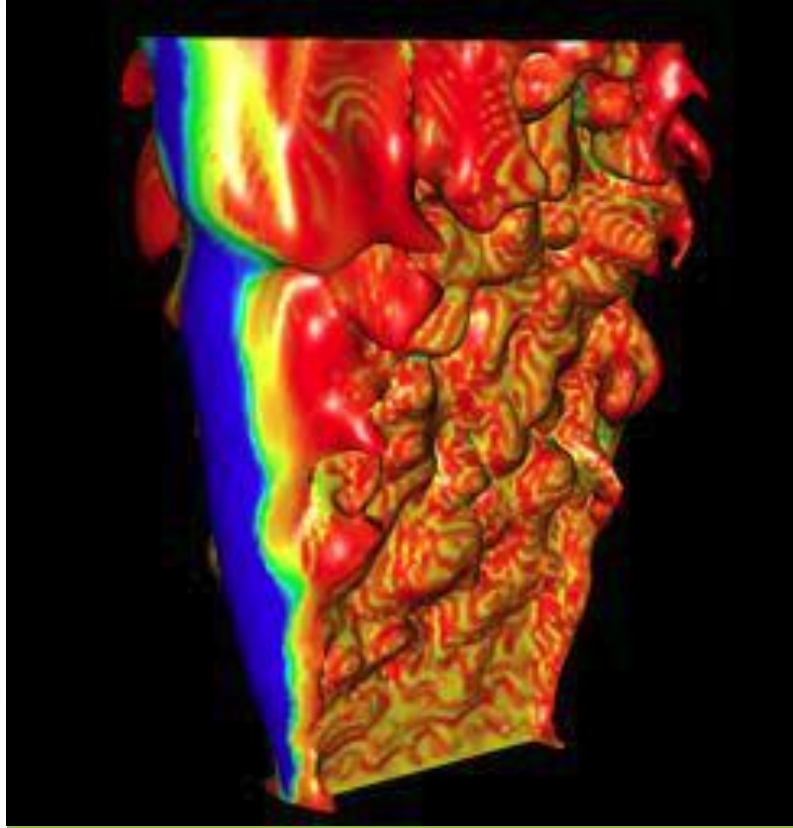
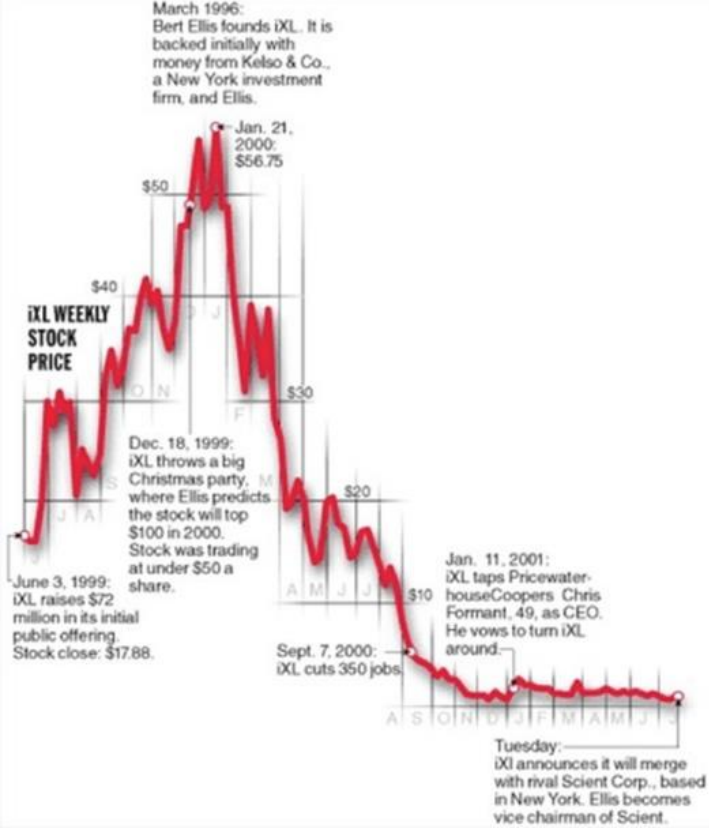
1 day 5 day 1 month 3 month 1 year 5 year



Open	112.71	Mkt cap	611.12B
High	113.05	P/E ratio	13.14
Low	112.28	Div yield	2.03%







**Information
Visualization**

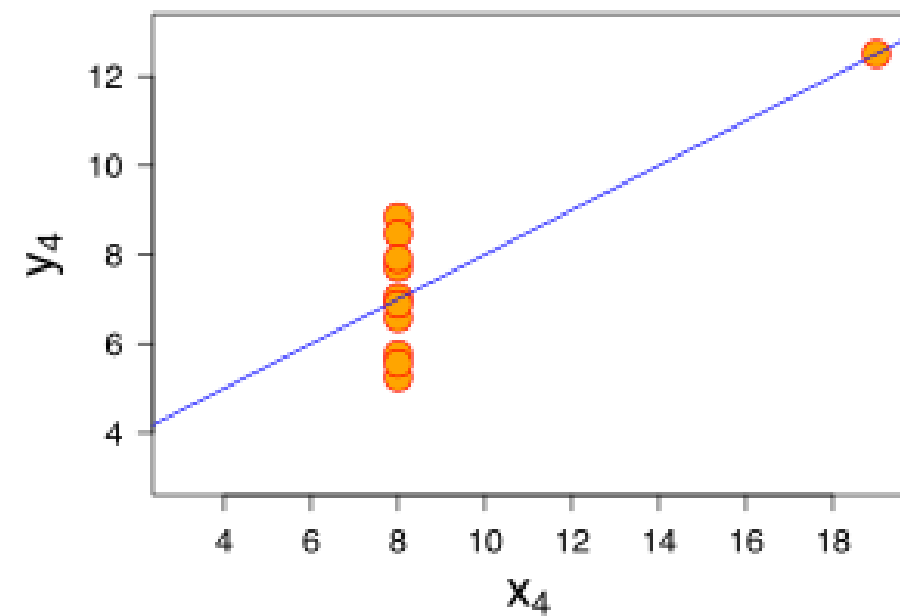
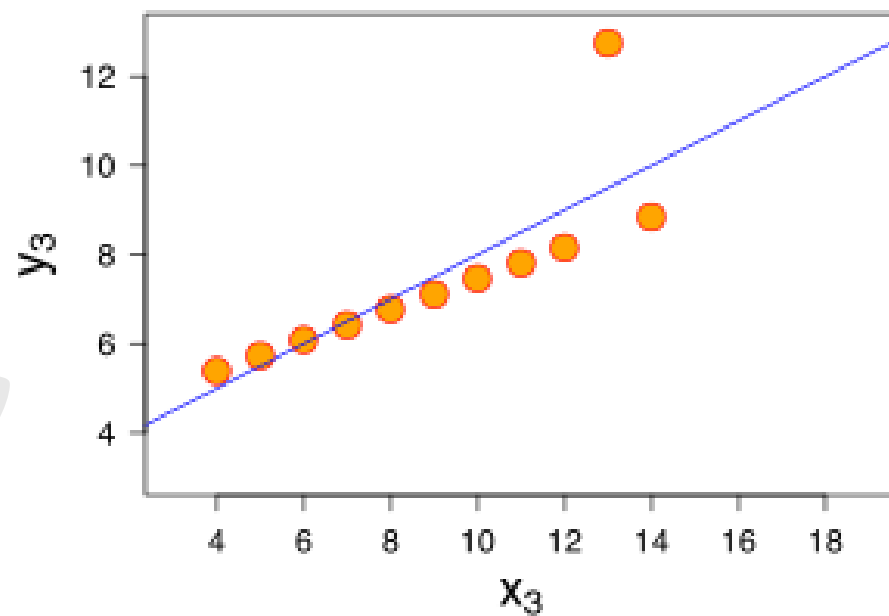
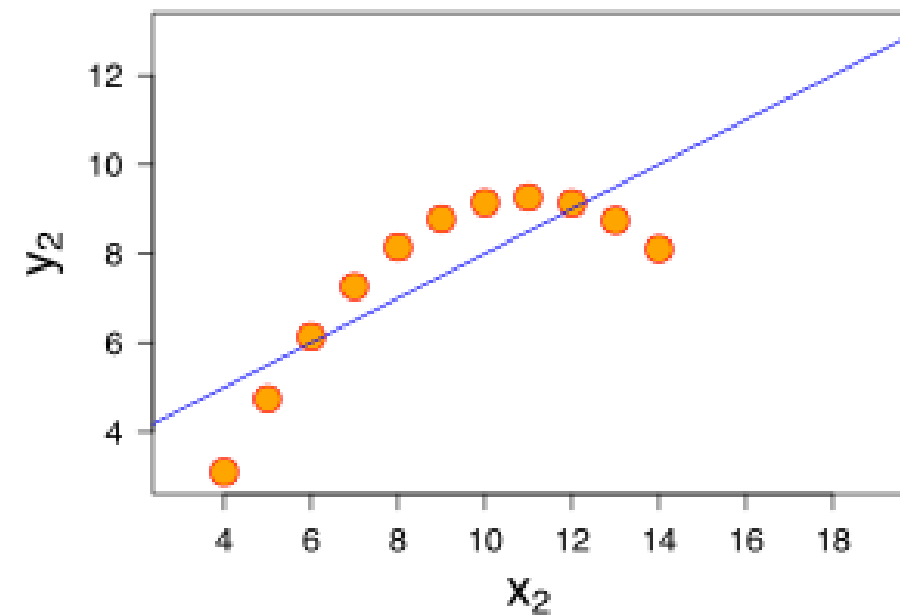
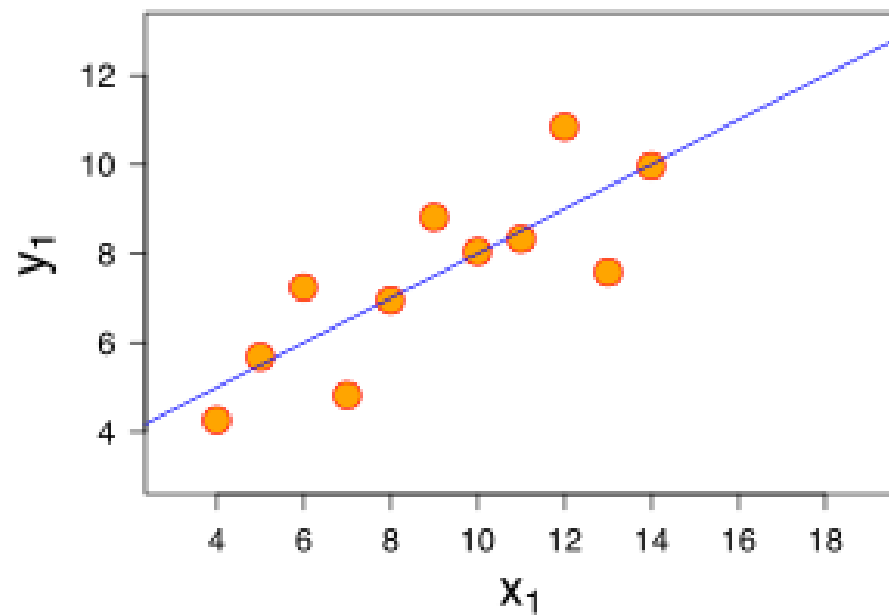
**Scientific
Visualization**

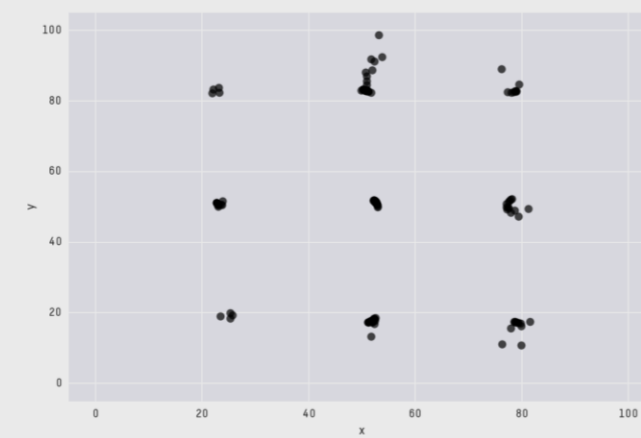
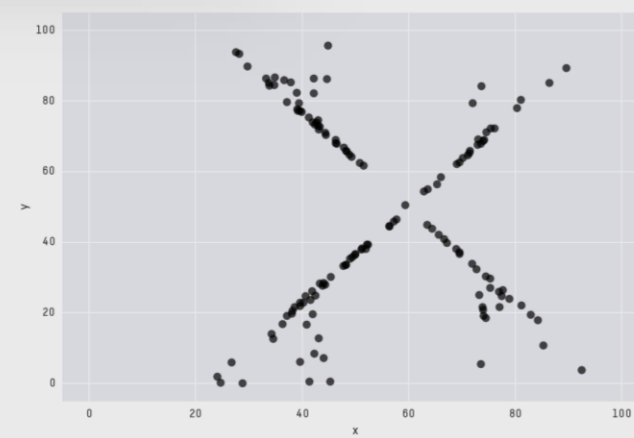
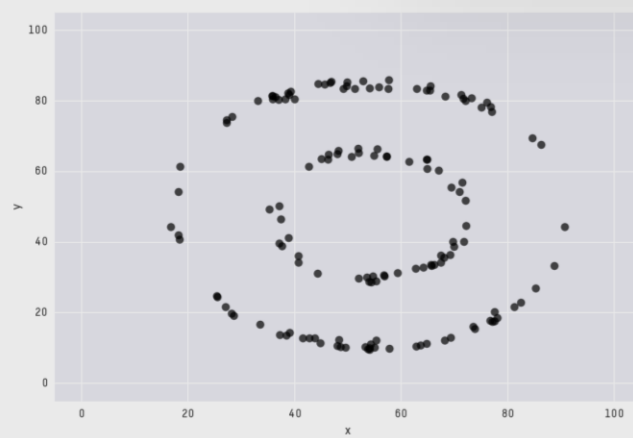
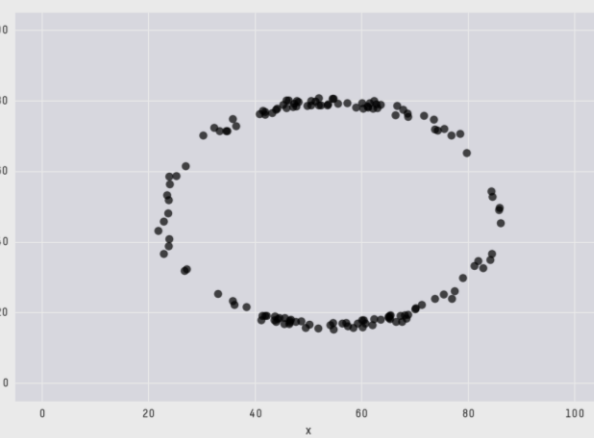
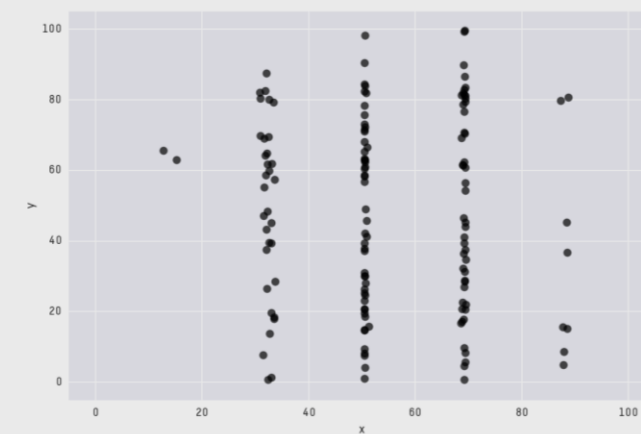
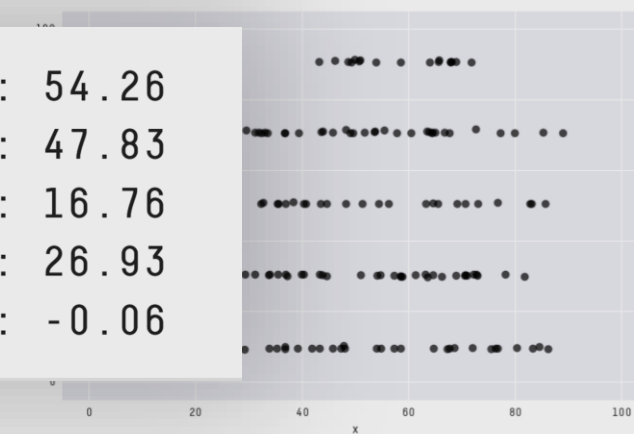
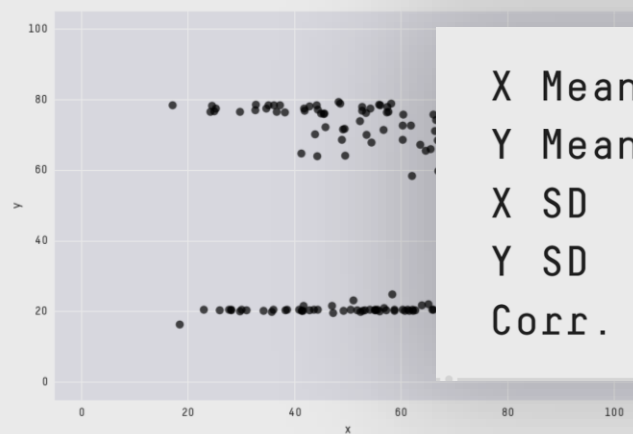
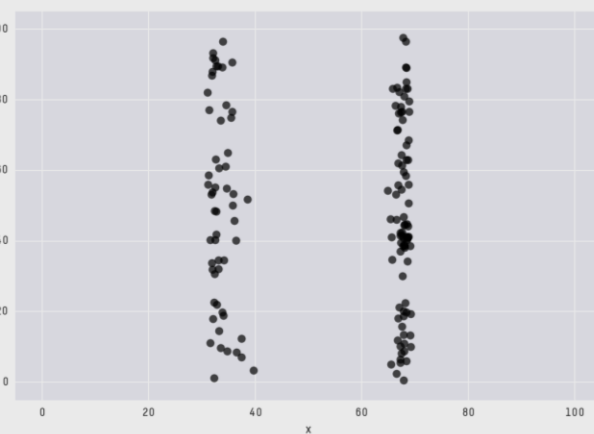
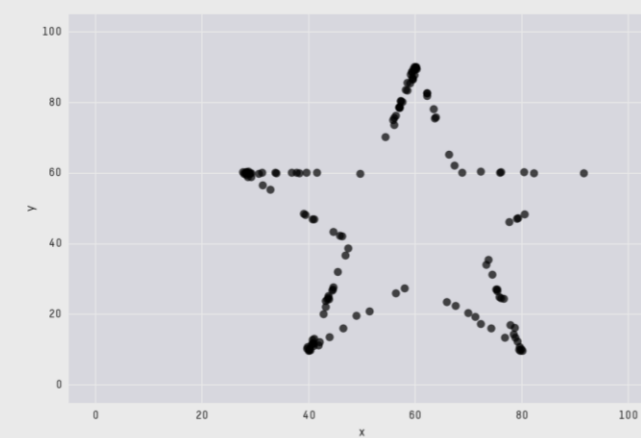
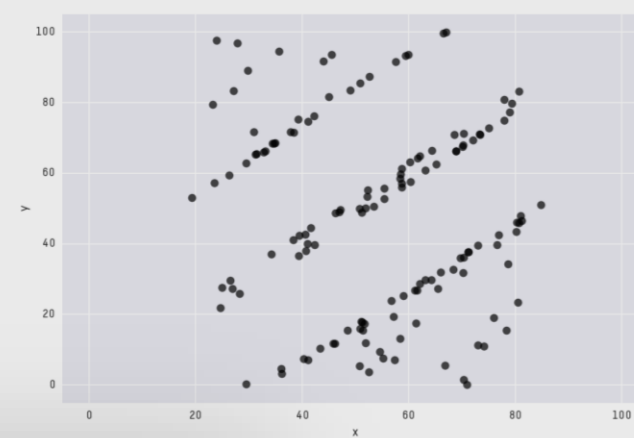
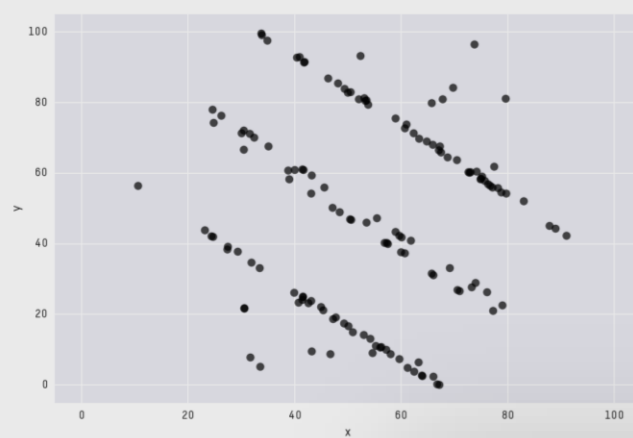
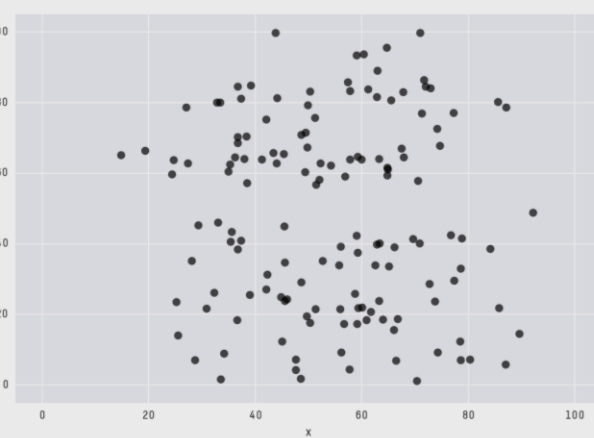
**Visual
Analytics**

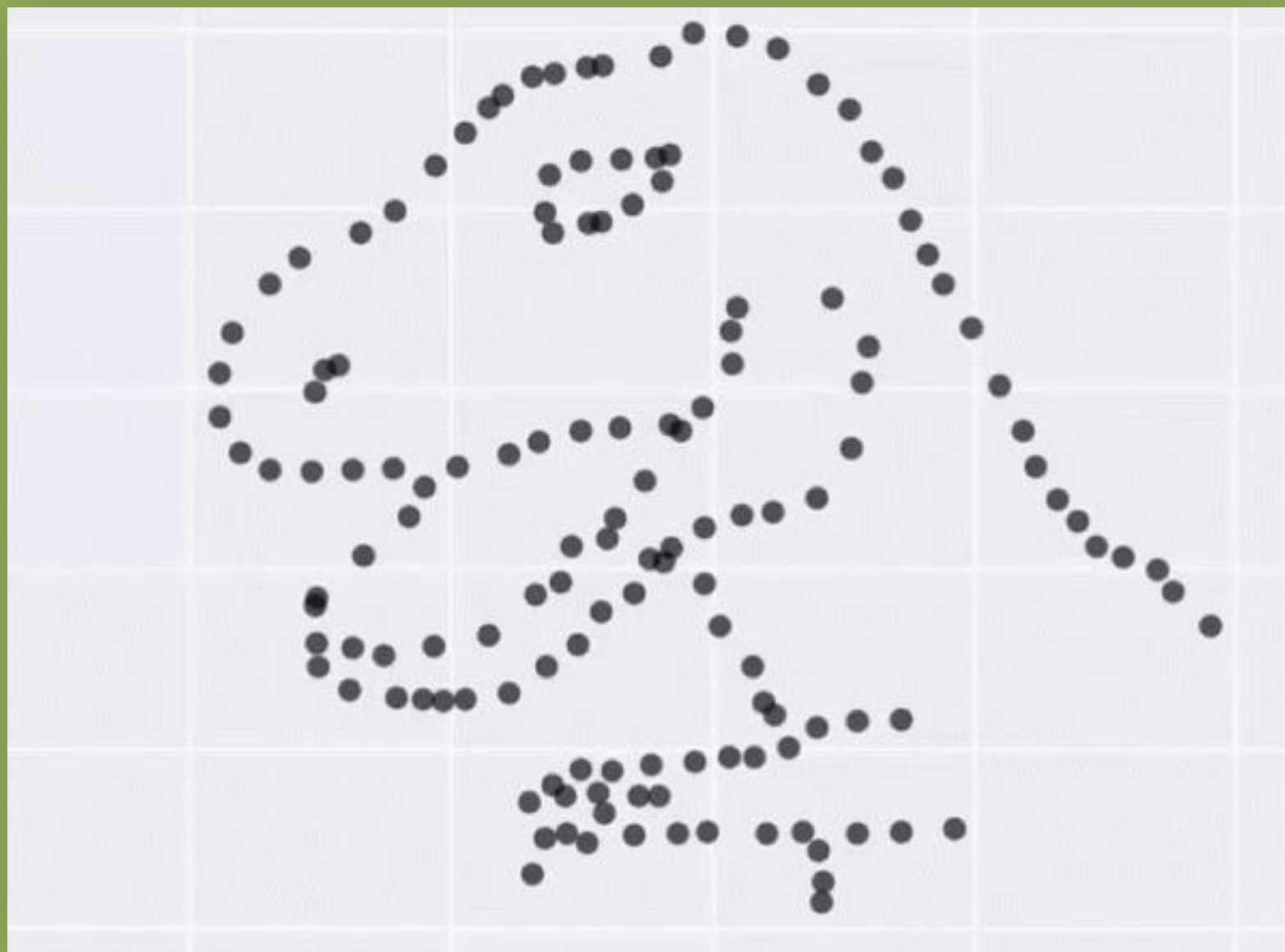
I		II		III		IV	
x	y	x	y	x	y	x	y
10	8,04	10	9,14	10	7,46	8	6,58
8	6,95	8	8,14	8	6,77	8	5,76
13	7,58	13	8,74	13	12,74	8	7,71
9	8,81	9	8,77	9	7,11	8	8,84
11	8,33	11	9,26	11	7,81	8	8,47
14	9,96	14	8,1	14	8,84	8	7,04
6	7,24	6	6,13	6	6,08	8	5,25
4	4,26	4	3,1	4	5,39	19	12,5
12	10,84	12	9,13	12	8,15	8	5,56
7	4,82	7	7,26	7	6,42	8	7,91
5	5,68	5	4,74	5	5,73	8	6,89

	I		II		III		IV	
	x	y	x	y	x	y	x	y
	10	8,04	10	9,14	10	7,46	8	6,58
	8	6,95	8	8,14	8	6,77	8	5,76
	13	7,58	13	8,74	13	12,74	8	7,71
	9	8,81	9	8,77	9	7,11	8	8,84
	11	8,33	11	9,26	11	7,81	8	8,47
	14	9,96	14	8,1	14	8,84	8	7,04
	6	7,24	6	6,13	6	6,08	8	5,25
	4	4,26	4	3,1	4	5,39	19	12,5
	12	10,84	12	9,13	12	8,15	8	5,56
	7	4,82	7	7,26	7	6,42	8	7,91
	5	5,68	5	4,74	5	5,73	8	6,89
SUM	99,00	82,51	99,00	82,51	99,00	82,50	99,00	82,51
AVG	9,00	7,50	9,00	7,50	9,00	7,50	9,00	7,50
STDEV	3,32	2,03	3,32	2,03	3,32	2,03	3,32	2,03

Anscombe's Quartet





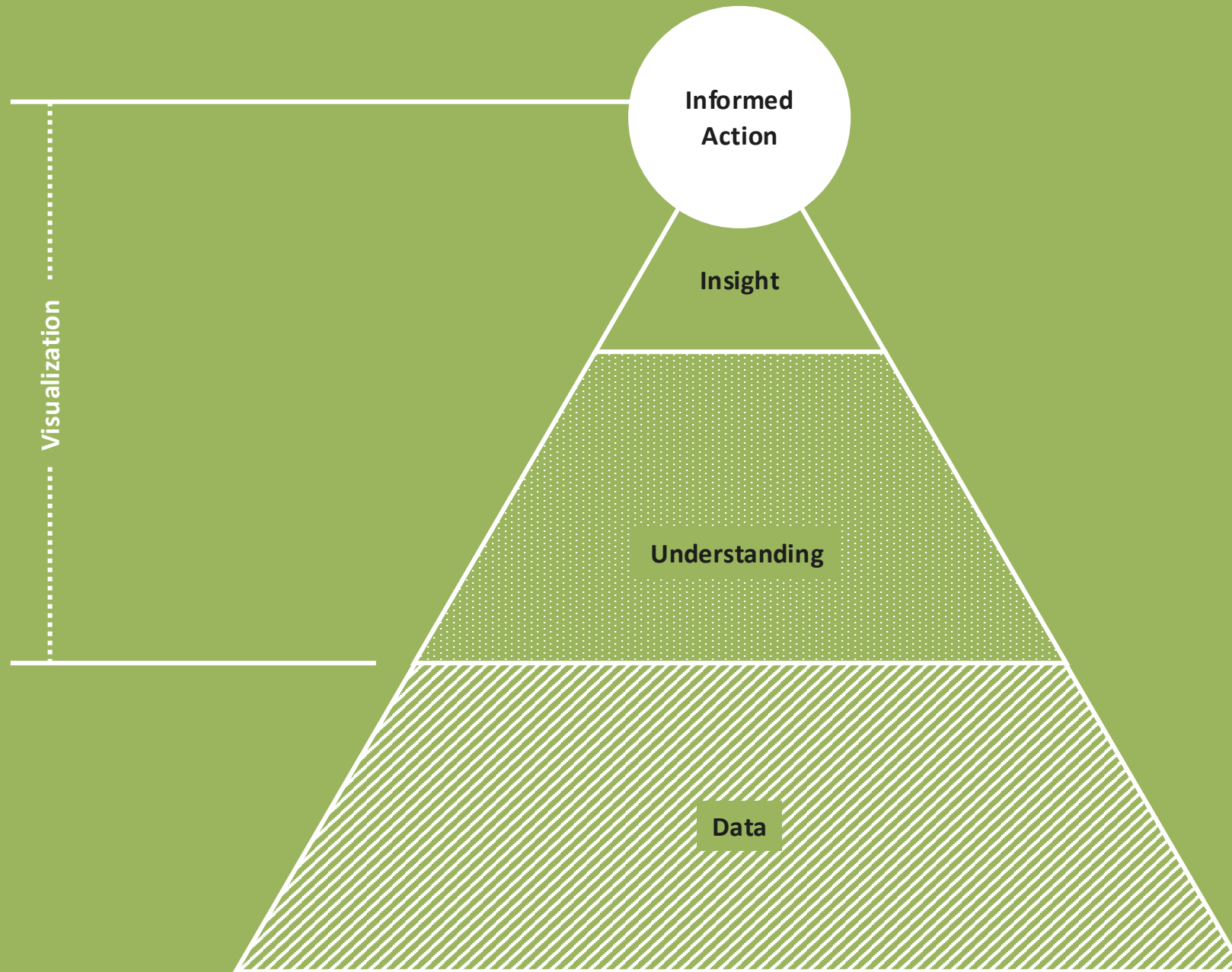








High-bandwidth
Subconscious
Parallel



**Creating information visualizations means
making tools for understanding data.**

Toolmaking is a craft.

**A craft consists of methods and techniques
relevant to a given medium.**

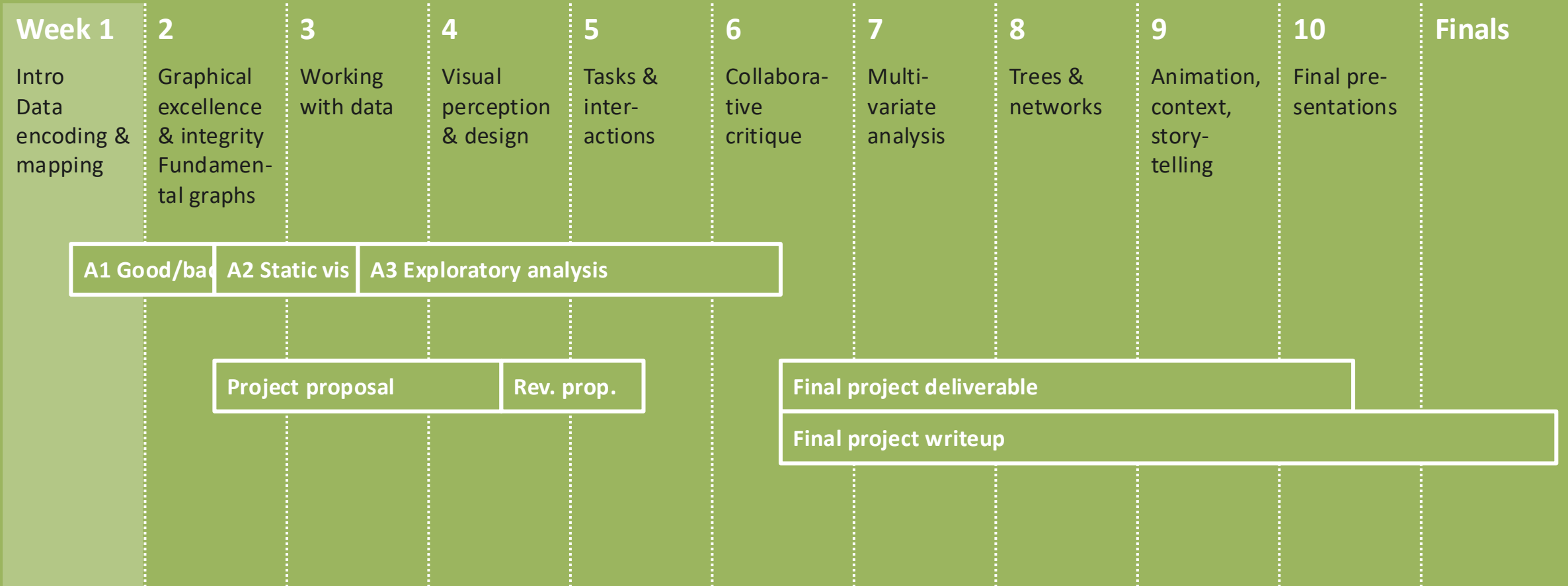
Syllabus

Canvas: <https://canvas.uw.edu/courses/1862679>

Thursdays

6-9:50p

SIG 233



40%

Main assignments

30%

Final project

20%

In-class participation

10%

Reading discussions

40%

Main assignments

30%

Final project

20%

In-class participation

10%

Reading discussions

1 Good and bad visualization

2 Static visualization critique

3 Exploratory data analysis

40%

Main assignments

30%

Final project

20%

In-class participation

10%

Reading discussions

- 1 Project proposal
- 2 Revised project proposal
- 3 Collaborative Critique
- 4 Final presentation
- 5 Final paper
- 6 Interactive visualization
- 7 Peer assessment form

40%

Main assignments

30%

Final project

20%

In-class participation

10%

Reading discussions

Work in groups

Multiple deliverables (proposal, presentations, paper)

Plan ahead, meet early and often

Data set choice is key

Technical excellence – balance between complexity of data and complexity of execution

Interview target users EARLY!

Doesn't have to be a dashboard/web app!

40%

Main assignments

30%

Final project

20%

In-class participation

10%

Reading discussions

Discussion contributions and assignments

Submit in-class assignments during or after class

If you miss a class, you cannot make up the in-class participation assignment

40%

Main assignments

30%

Final project

20%

In-class participation

10%

Reading discussions

1 required reading per week

Optional readings

Graded discussion posting

One question to be addressed per week

Break

Data encoding & mapping



High-bandwidth
Subconscious
Parallel

123
ABC



123
ABC



Nominal

Apples
Oranges
Bananas

Jay
Pat
Chris

Red
Green
Blue

Ordinal

First
Second
Third

Grade A
Grade AA
Grade B

Small
Medium
Large

Quantitative

Interval —

Jan 12th 2022
Jan 19th 2022
Jan 26th 2022

1 10%
2 25%
3 50%

————— Ratio

123
ABC



Nominal

= ne (not equal)

Ordinal

= ne < > <= >=

Quantitative

Interval

= ne < > <= >= - (distance) /

 Ratio

123
ABC



Nominal

Can you tell the difference?

Ordinal

Is there an order?

Quantitative

Can you measure the difference?

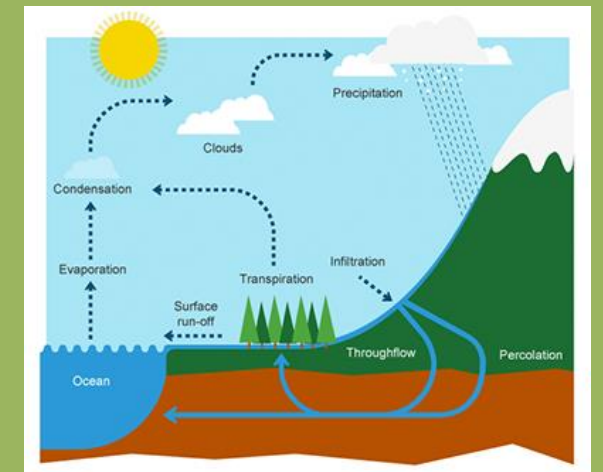
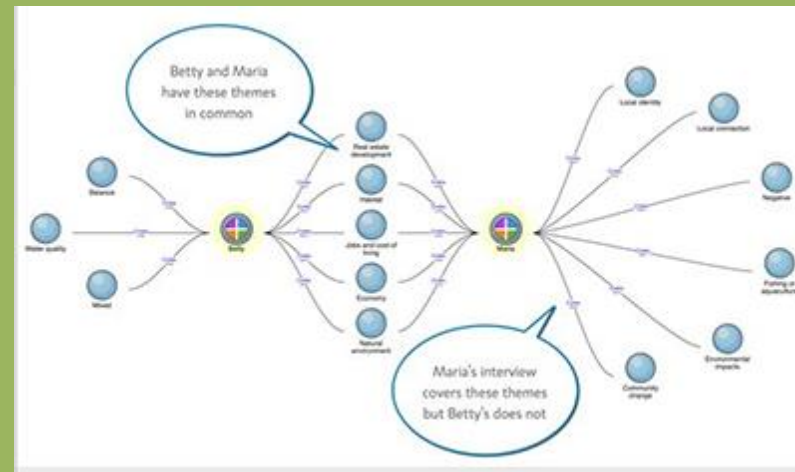
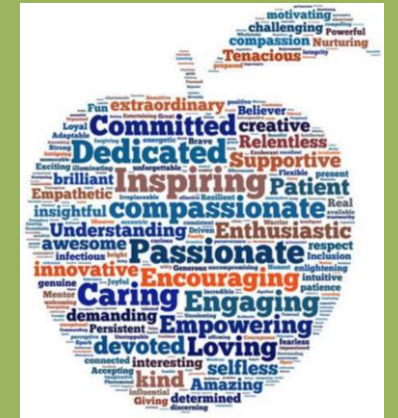
123
ABC



Nominal?

Ordinal?

Quantitative?



Exercise

Model	Origin	Year	Cylinders	Horsepower	MPG	Weight
amc ambassador dpl	US	70	8	190	15	3,850
chevrolet vega (sw)	US	71	4	72	22	2,408
chevrolet vega	US	72	4	90	20	2,408
ford pinto runabout	US	72	4	86	21	2,226
mazda rx2 coupe	Japan	72	3	97	19	2,330
amc matador (sw)	US	72	8	150	15	3,892
ford pinto (sw)	US	72	4	86	22	2,395
dodge coronet custom	US	73	8	150	15	3,777
toyota carina	Japan	73	4	88	20	2,279
chevrolet vega	US	73	4	72	21	2,401
datsum 610	Japan	73	4	94	22	2,379
ford pinto	US	73	4	85	19	2,310
chevrolet chevelle malibu classic	US	74	6	100	16	3,781
buick century	US	75	6	110	17	3,907
chevroelt chevelle malibu	US	75	6	105	16	3,897
plymouth fury	US	75	6	95	18	3,785
amc matador	US	76	8	120	15.5	3,962
mercedes-benz 280s	Europe	76	6	120	16.5	3,820
chevrolet caprice classic	US	77	8	145	17.5	3,880
chevrolet caprice classic	US	79	8	130	17	3,840
mercury grand marquis	US	79	8	138	16.5	3,955
dodge st. regis	US	79	8	135	18.2	3,830
vw rabbit	Europe	80	4	76	41.5	2,144
mazda glc	Japan	80	4	65	46.6	2,110
vw pickup	Europe	82	4	52	44	2,130

Can you tell the difference?

Is there an order?

Can you measure the difference?

123
ABC

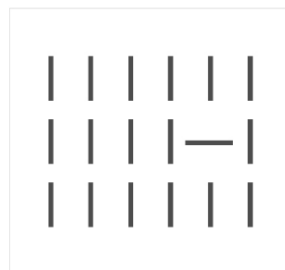




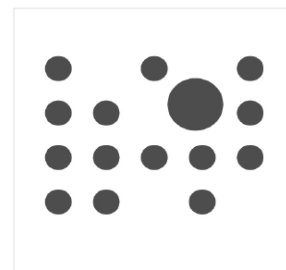
Length



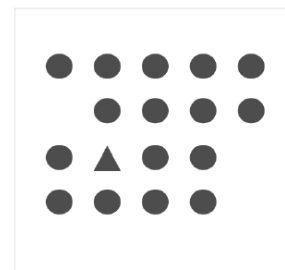
Width



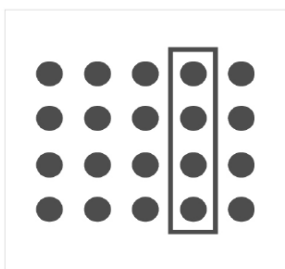
Orientation



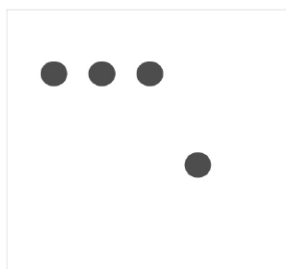
Size



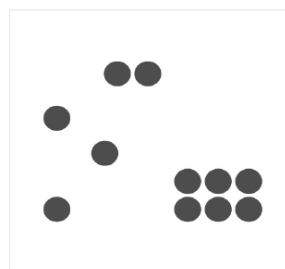
Shape



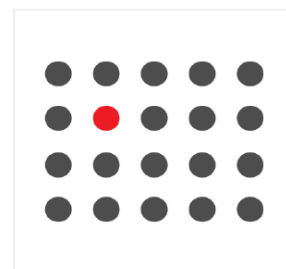
Enclosure



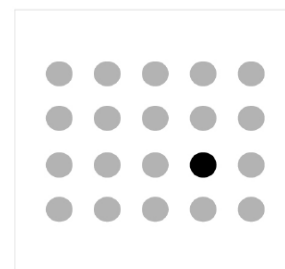
Position



Grouping



Color Hue



Color
Intensity

Expressiveness

A set of facts is expressible in a visual language if the sentences (i.e. the visualizations) in the language express all the facts in the set of data, and only the facts in the data.

Effectiveness

A visualization is more effective than another visualization if the information conveyed by one visualization is more readily perceived than the information in the other visualization.

Expressiveness

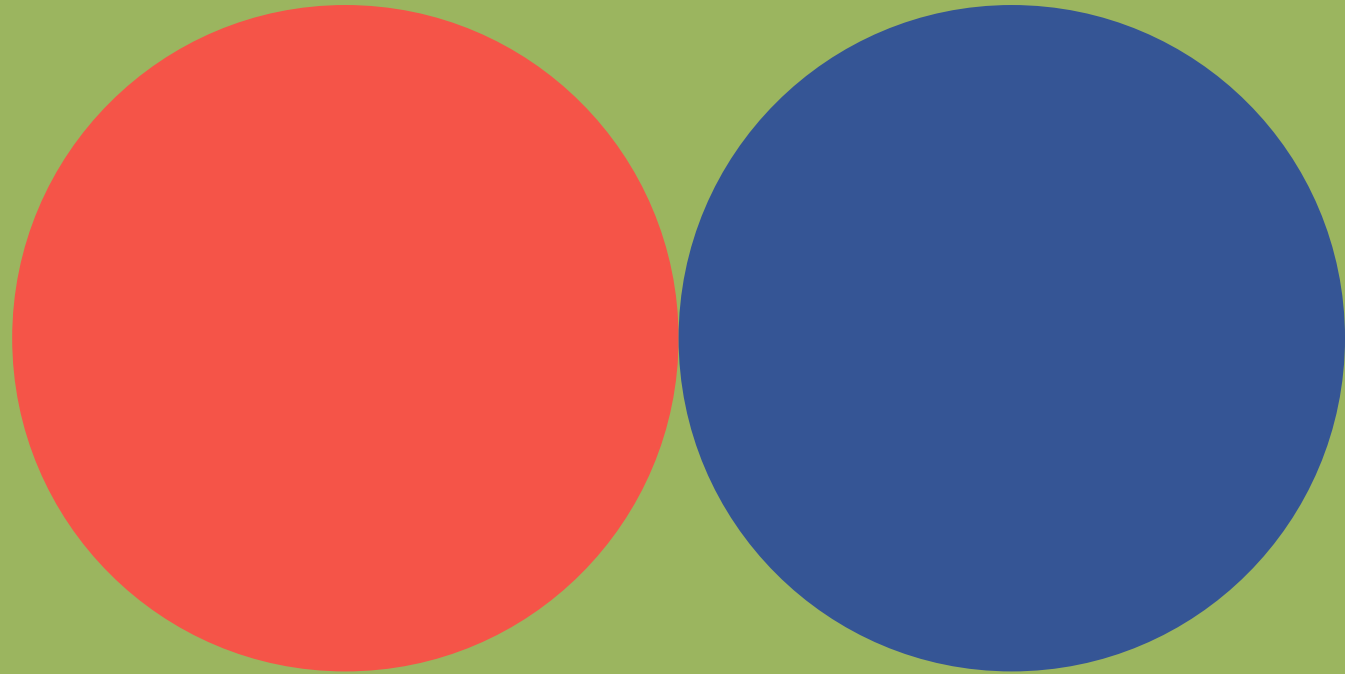
A set of facts is expressible in a visual language if the sentences (i.e. the visualizations) in the language express all the facts in the set of data, and only the facts in the data.

Effectiveness

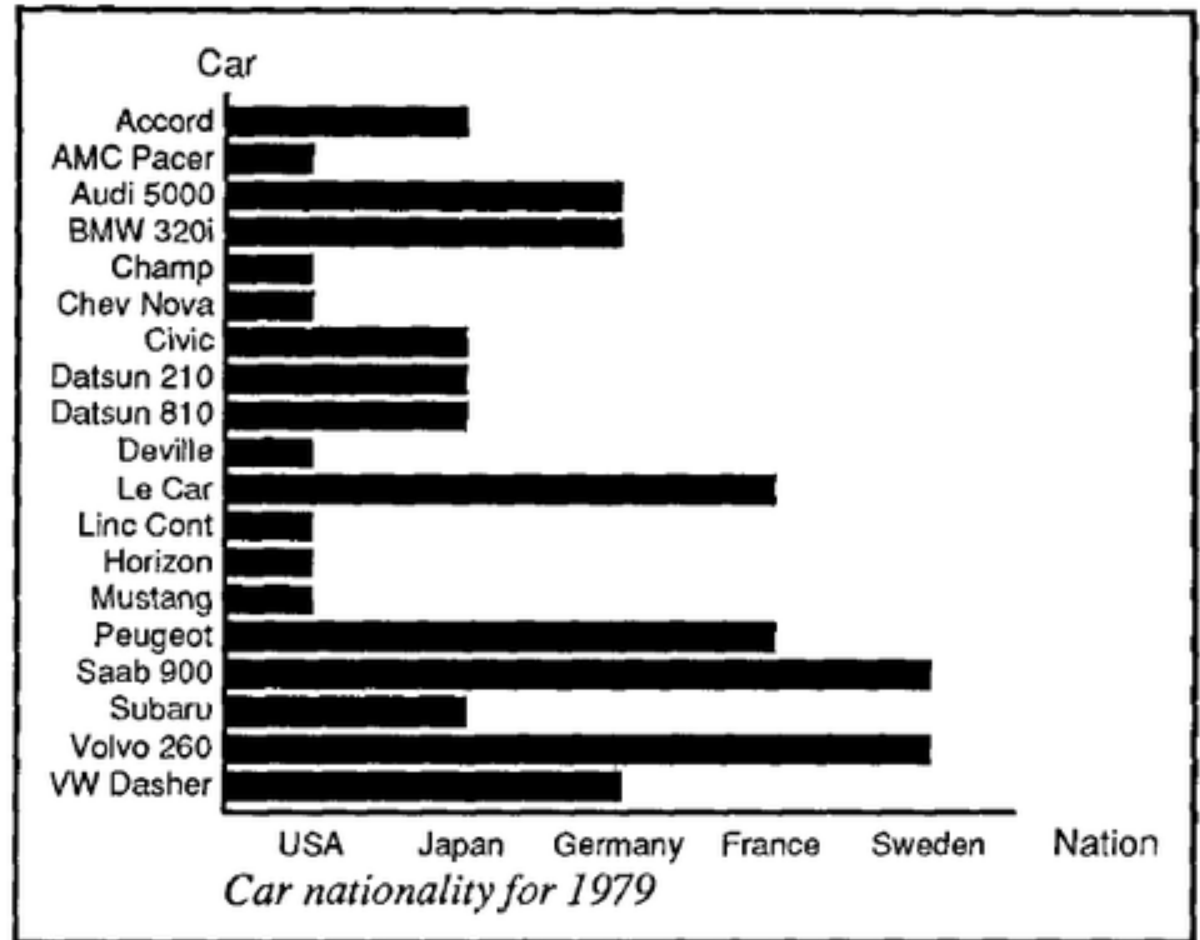
A visualization is more effective than another visualization if the information conveyed by one visualization is more readily perceived than the information in the other visualization.

**Can't express
the facts.**

**Which color is
greater than the
other?**



Expressing facts
not in the data.



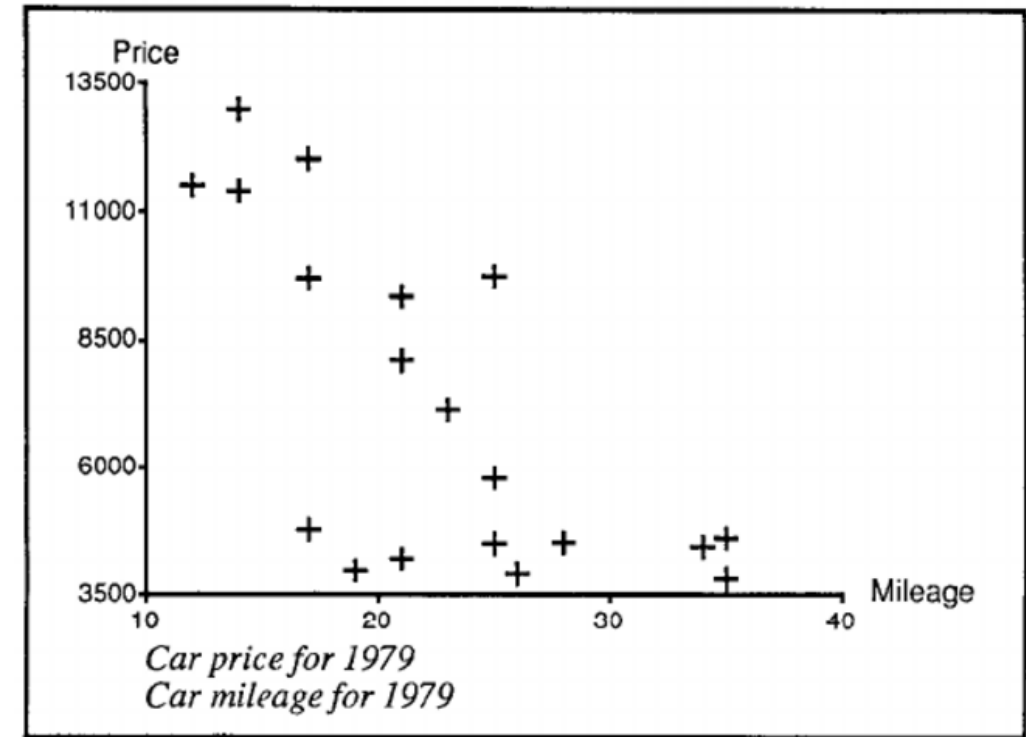
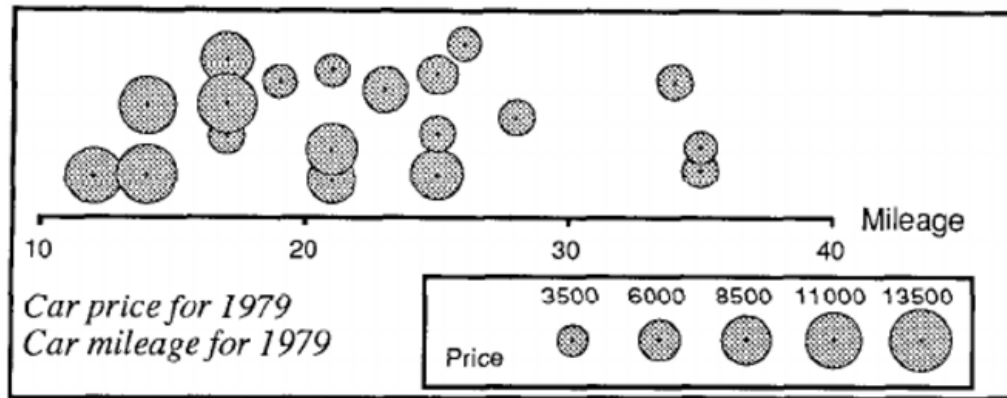
Expressiveness

A set of facts is expressible in a visual language if the sentences (i.e. the visualizations) in the language express all the facts in the set of data, and only the facts in the data.

Effectiveness

A visualization is more **effective** than another visualization if the information conveyed by one visualization is more readily perceived than the information in the other visualization.

Which is a more effective visualization of the data?



123
ABC



Nominal

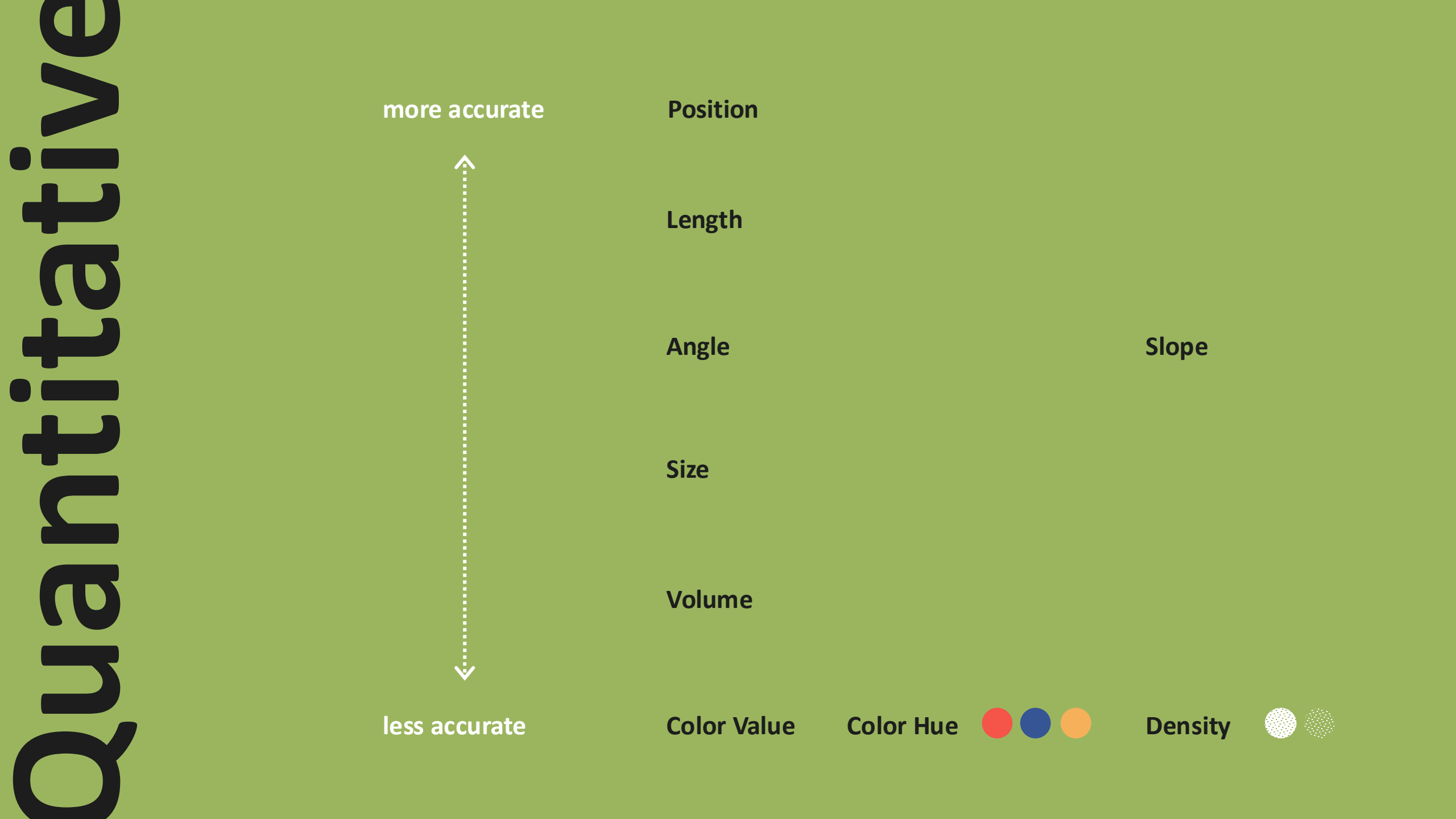
Can you tell the difference?

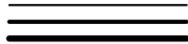
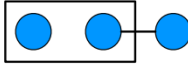
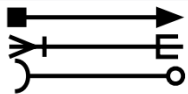
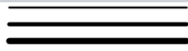
Ordinal

Is there an order?

Quantitative

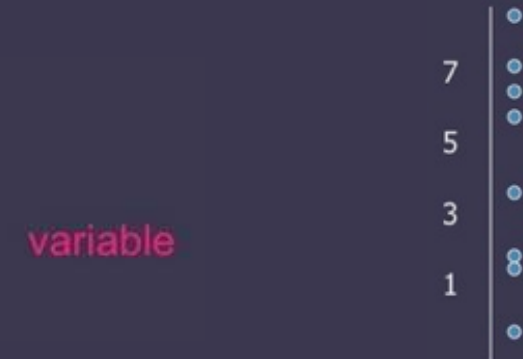
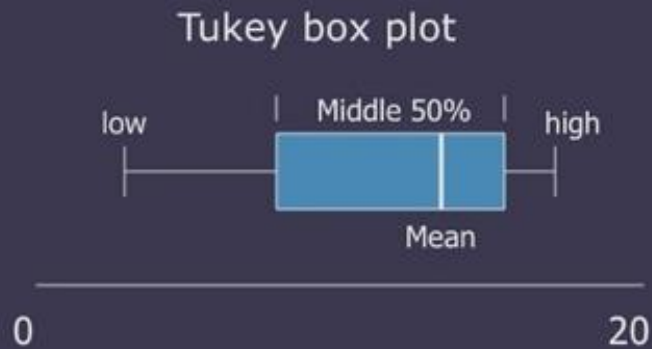
Can you measure the difference?



<u>Example</u>	<u>Encoding</u>	<u>Ordered</u>	<u>Useful values</u>	<u>Quantitative</u>	<u>Ordinal</u>	<u>Nominal</u>	<u>Relational</u>
	position, placement	yes	infinite	Good	Good	Good	Good
1, 2, 3; A, B, C	text labels	optional (alphabetical or numbered)	infinite	Good	Good	Good	Good
	length	yes	many	Good	Good		
	size, area	yes	many	Good	Good		
	angle	yes	medium/few	Good	Good		
	pattern density	yes	few	Good	Good		
	weight, boldness	yes	few		Good		
	saturation, brightness	yes	few		Good		
	color	no	few (< 20)			Good	
	shape, icon	no	medium			Good	
	pattern texture	no	medium			Good	
	enclosure, connection	no	infinite			Good	Good
	line pattern	no	few				Good
	line endings	no	few				Good
	line weight	yes	few		Good		

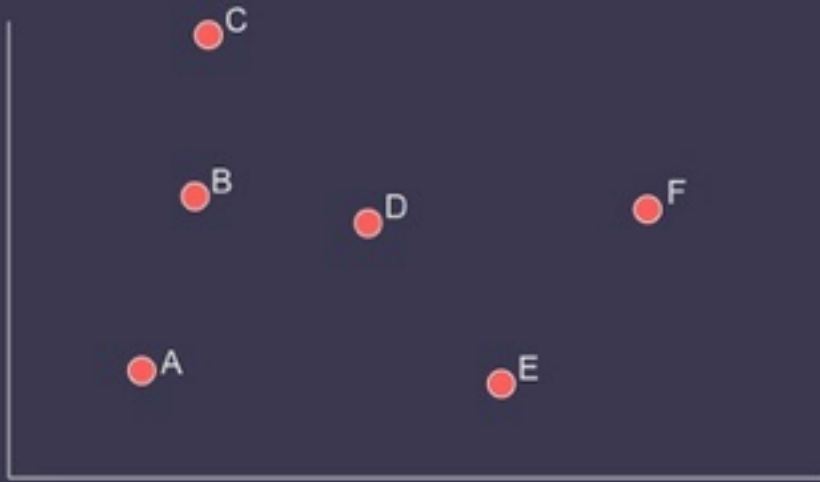
Univariate Data

1 Variable



Bivariate Data

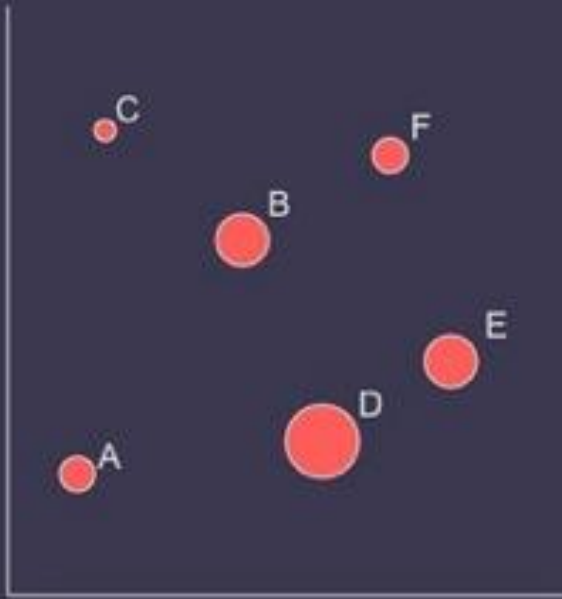
2 Variables



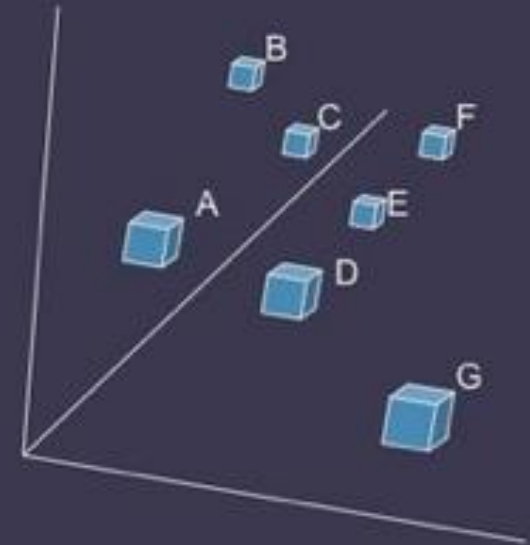
	A	B	C
1			
2			

Trivariate Data

3 Variables



	A	B	C
1			
2			
3			





Points (0D)



Lines (1D)



Areas (2D)



Volume (3D)

Marks

Exercise

Model	Origin	Year	Cylinders	Horsepower	MPG	Weight
amc ambassador dpl	US	70	8	190	15	3,850
chevrolet vega (sw)	US	71	4	72	22	2,408
chevrolet vega	US	72	4	90	20	2,408
ford pinto runabout	US	72	4	86	21	2,226
mazda rx2 coupe	Japan	72	3	97	19	2,330
amc matador (sw)	US	72	8	150	15	3,892
ford pinto (sw)	US	72	4	86	22	2,395
dodge coronet custom	US	73	8	150	15	3,777
toyota carina	Japan	73	4	88	20	2,279
chevrolet vega	US	73	4	72	21	2,401
datsum 610	Japan	73	4	94	22	2,379
ford pinto	US	73	4	85	19	2,310
chevrolet chevelle malibu classic	US	74	6	100	16	3,781
buick century	US	75	6	110	17	3,907
chevroelt chevelle malibu	US	75	6	105	16	3,897
plymouth fury	US	75	6	95	18	3,785
amc matador	US	76	8	120	15.5	3,962
mercedes-benz 280s	Europe	76	6	120	16.5	3,820
chevrolet caprice classic	US	77	8	145	17.5	3,880
chevrolet caprice classic	US	79	8	130	17	3,840
mercury grand marquis	US	79	8	138	16.5	3,955
dodge st. regis	US	79	8	135	18.2	3,830
vw rabbit	Europe	80	4	76	41.5	2,144
mazda glc	Japan	80	4	65	46.6	2,110
vw pickup	Europe	82	4	52	44	2,130

Survey

**We want to make sure we
create balanced teams for
the final project.**

<https://forms.gle/jKD99n3UNtUxbRx9A>

Due by tonight!

Assignment

1

Good

Static visualization

Intended for a general audience

Not from papers or textbooks on data visualization

and not previously analyzed

Describe why

Due 11:59p Tuesday 14 Jan

1

Bad